

Project Code: B1082301
Project Number: TANK 3.0 (SEALION)
PCB Number: 23X22-SA
PCB Version: SA
PCB Size: TBU
PCB Layer: 8
PCB Thickness: 1.57mm

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NOPOP=no populated

562R2F-GP
562 = 562 ohm, (2K2R means 2.2K ohm)
2 = size 0402
F = 1% tolerance
GP= Green Part (RoHS)
Wiwynn RC size code as below:
1 = 0201
2= 0402
3= 0603
5= 0805
6= 1206
Wiwynn R tolerance code as below:
D=0.5%
F= 1%
J= 5%

SCD1U10V2KX-5GP
D1U = 0.1uF (2D2U means 2.2uF)
10Voltage (6D3V means 6.3V)
2 = size 0402, K tolerance
K=tolerance
[Wiwynn C code as below:]
G=2%
J=5%
K=10%
M=20%
X=temp characteristics
[(RoHS) Wiwynn C Series/Temp]
N=NPO
X=X7R/X5R
Y=Y5V
-5=different symbol/customer
GP= Green Part (RoHS)

SYSTEM BLOCK DIAGRAM

<Variant Name>



Wiwynn Corporation
8F, 90, Sec.1, Xintai 5th Rd.
New Taipei City 22102, Taiwan (R.O.C.)
<http://www.wiwynn.com>

Title003_ Block Diagram

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TERMINAL BOARD DIAGRAM

<Variant Name>



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<http://www.wiwynn.com>

Title

004 TERMINAL BOARD DIAGRAM

Size

A3

Document Number

Sealion

Rev

SA

Date

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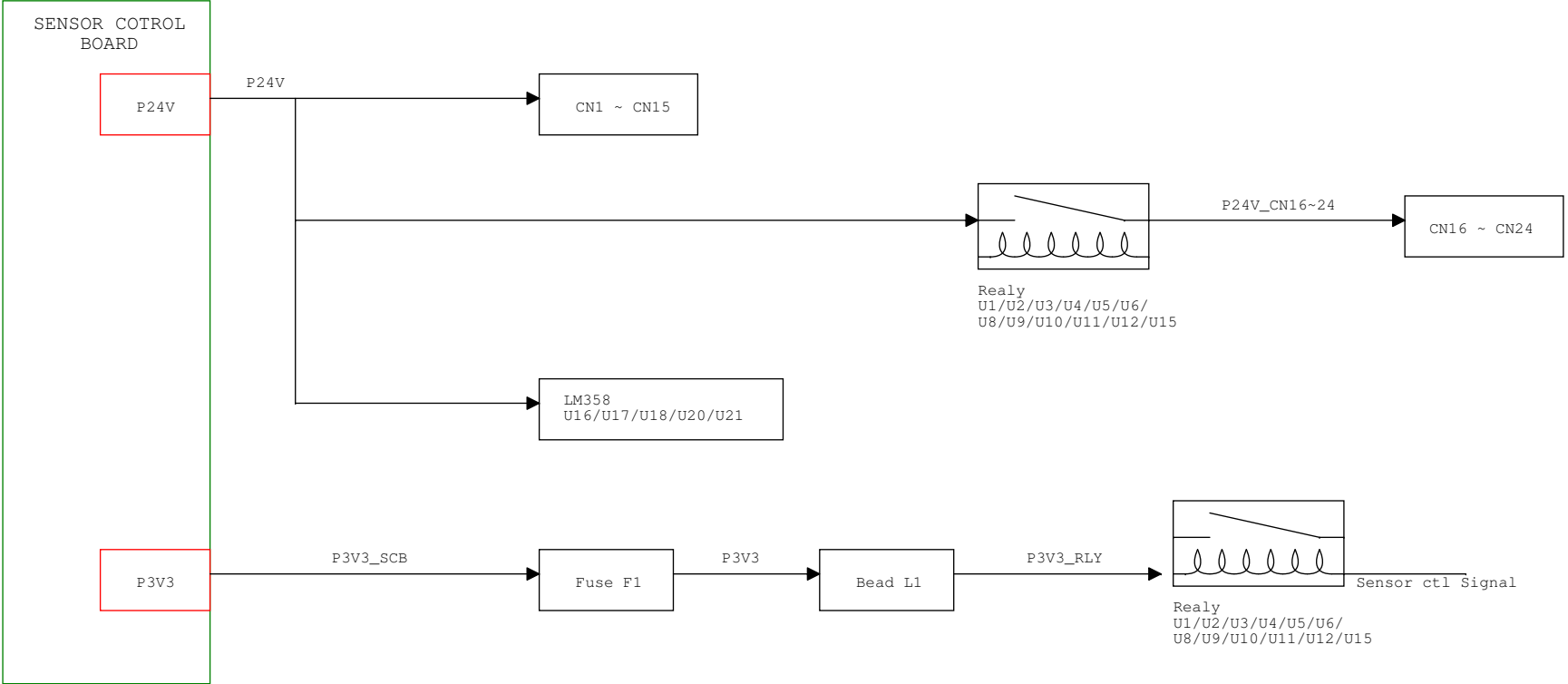
Sheet

4

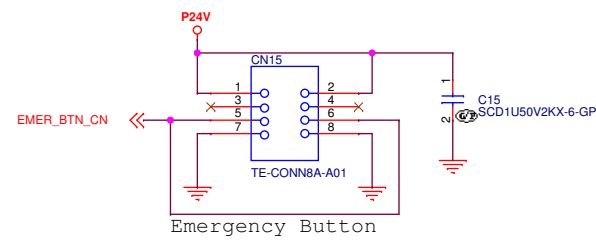
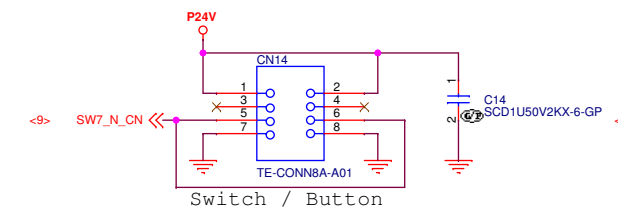
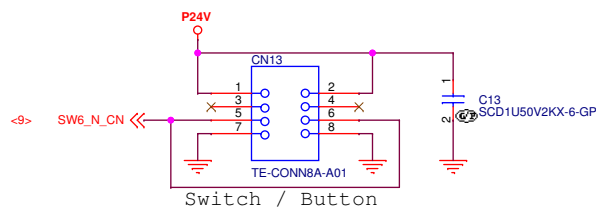
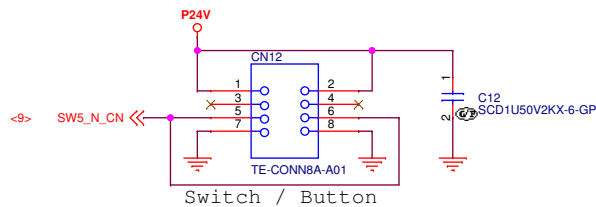
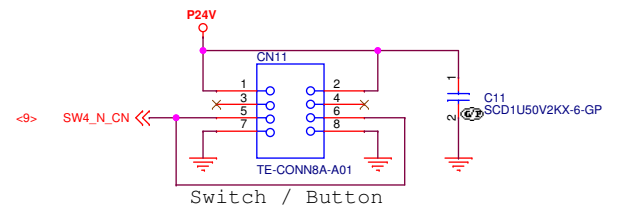
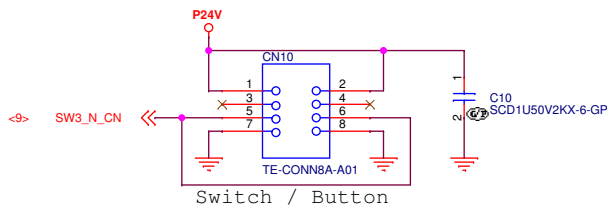
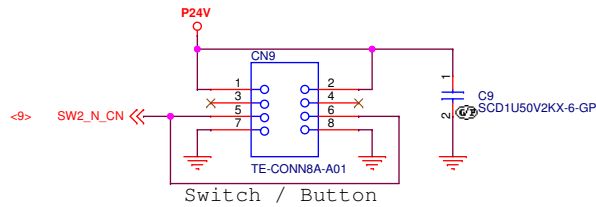
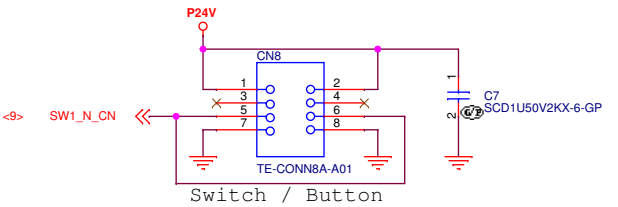
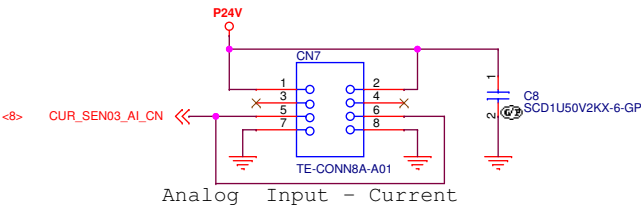
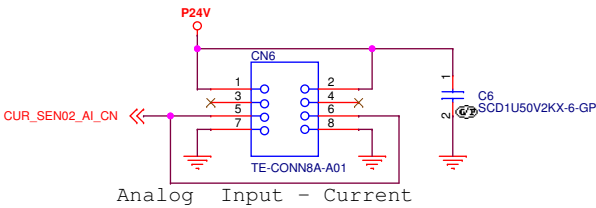
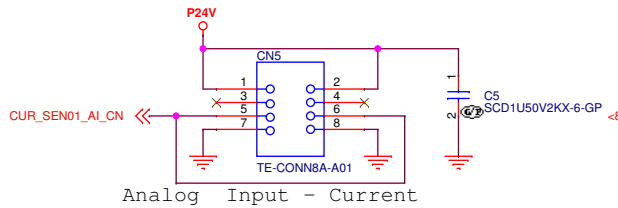
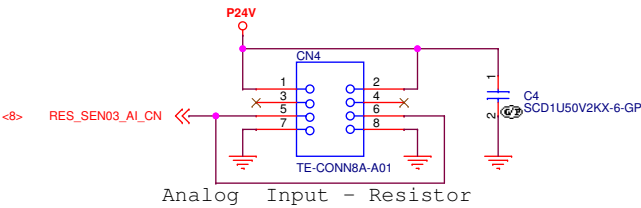
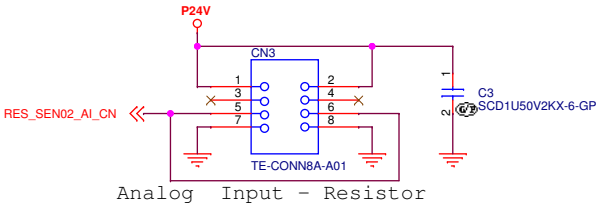
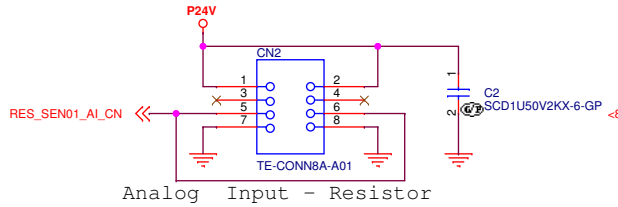
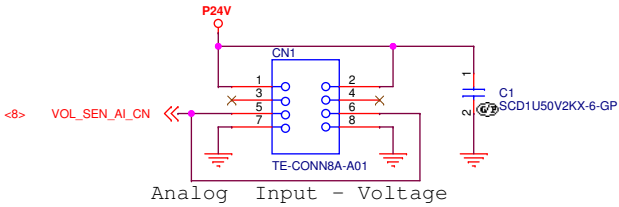
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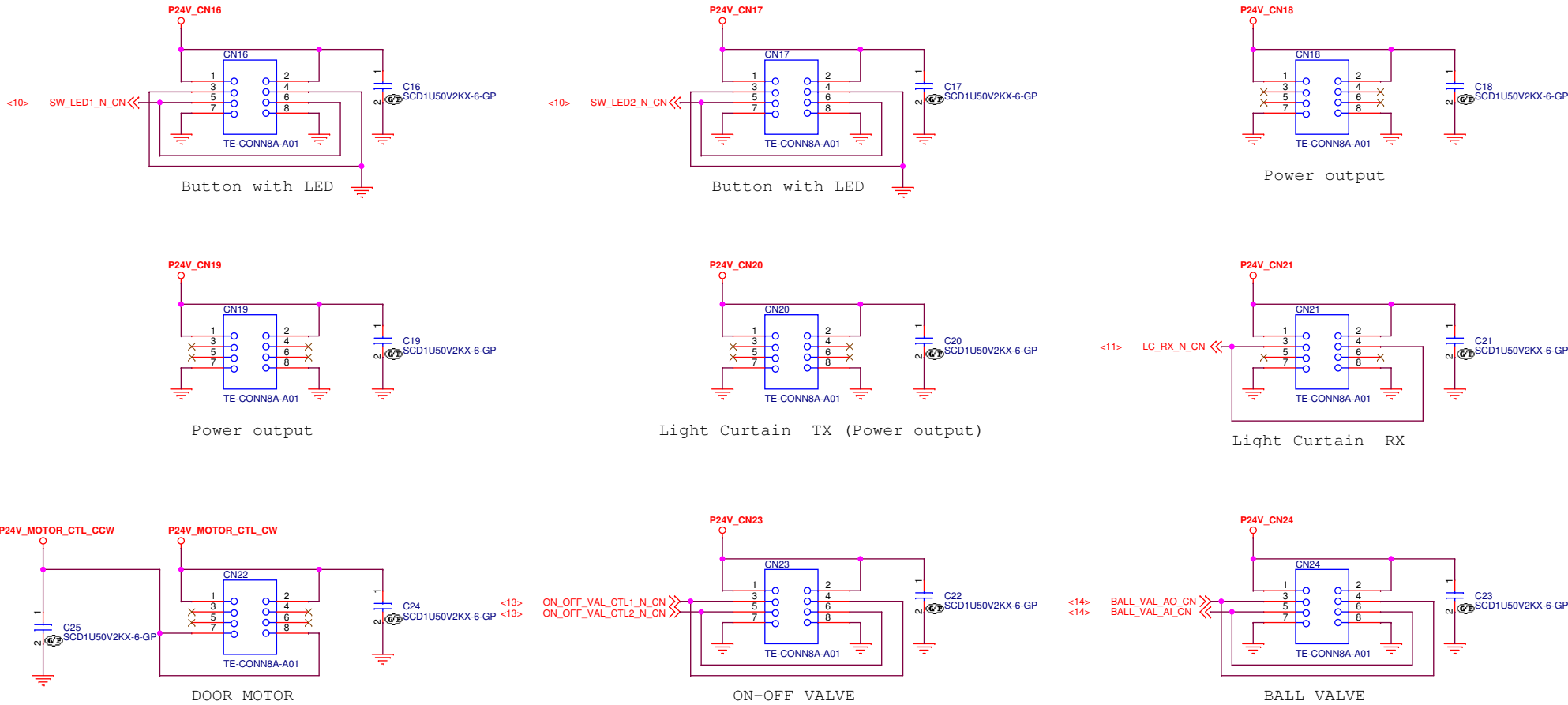
POWER DIAGRAM (TBU)



Terminal CONN CN1 ~ CN15



Terminal CONN CN16 ~ CN24

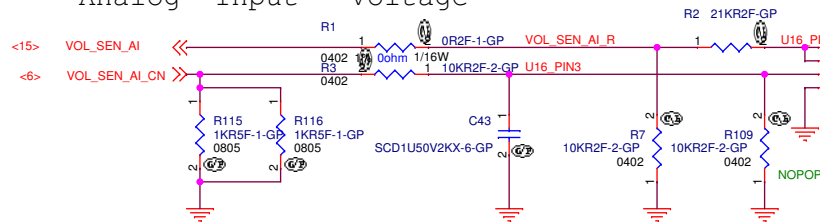


FUSE for Relay

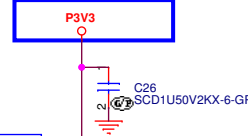


Sensor IO (CN1~CN7)

CN1 : Humidity sensor
Analog Input - Voltage

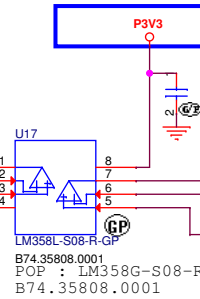
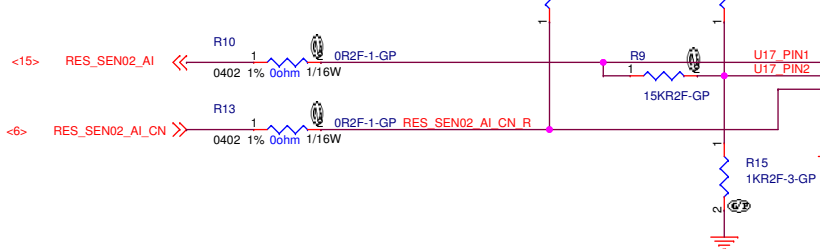


20231023:
Change U16 and U17 power from P24V to P3V3

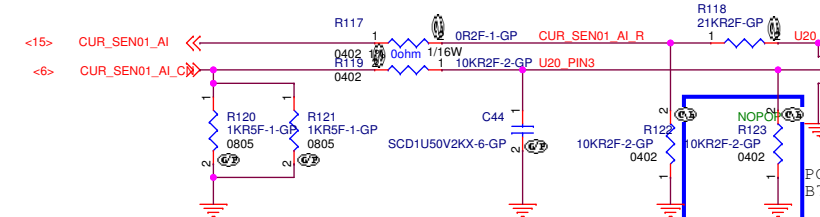


CN2 : Temp sensor
Analog Input - Resistor

CN3 : Temp sensor
Analog Input - Resistor



CN4 : Temp sensor
Analog Input - Resistor



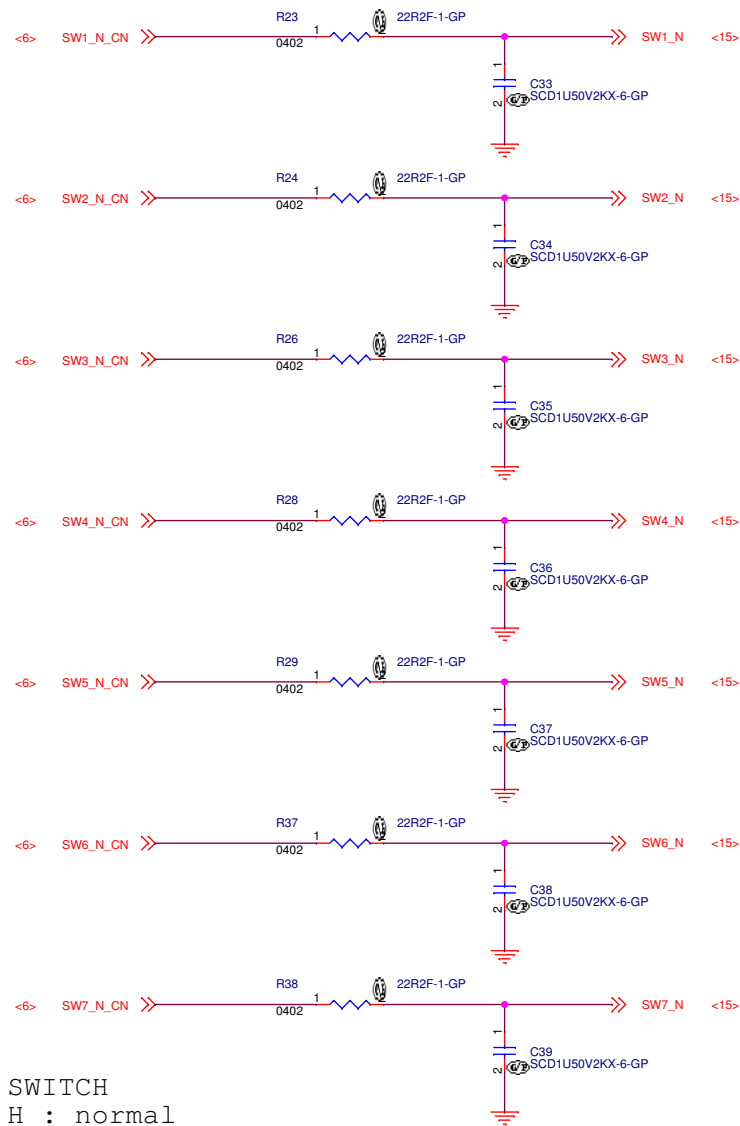
20231023:
Change R123 to NOPOP

CN5 ~ CN7 : Pressure sensor / Flow sensor / LiqSensor
Analog input -current (4 ~ 20 mA)
Need to transfer to voltage

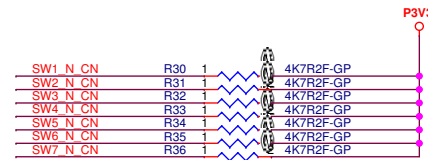
<Variant Name>

wiwynn		Wiwynn Corporation 8F, 90, Sec.1, Xintai 5th Rd. New Taipei City 22102, Taiwan (R.O.C.) http://www.wiwynn.com	
Title		008 Sensor IO (CN1~CN7)	
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SWITCH / EMER BUTTON (CN8~CN15)

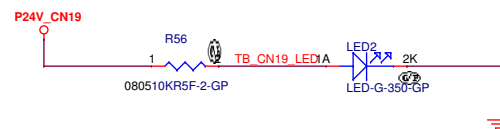
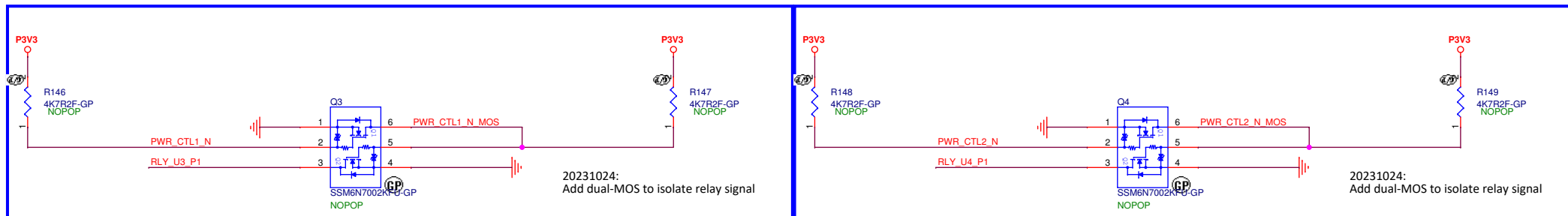
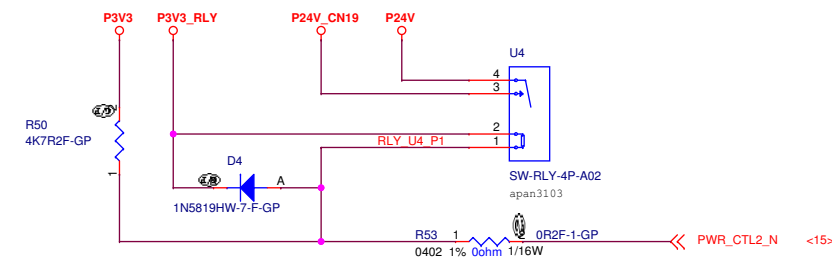
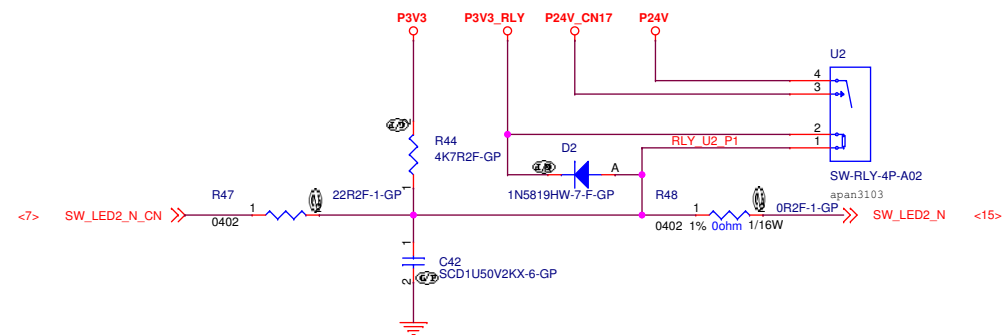


PU RES for Switch (CN8 ~ CN14)

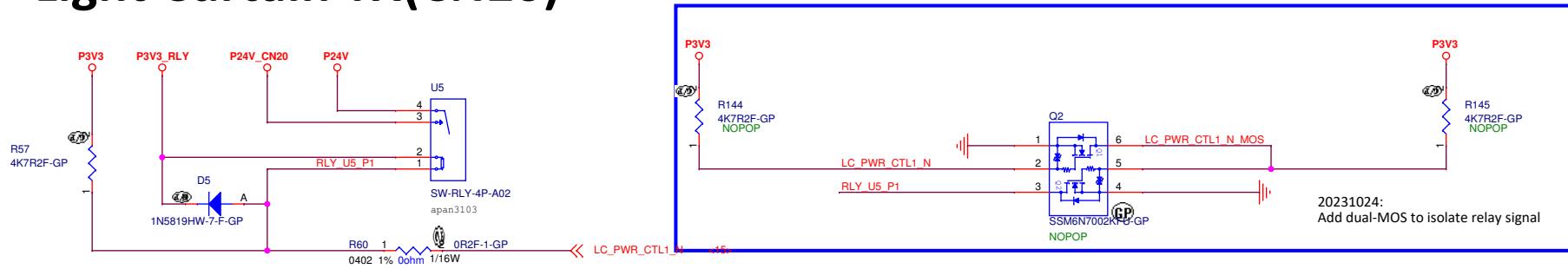


SWITCH
H : normal
L : btn press

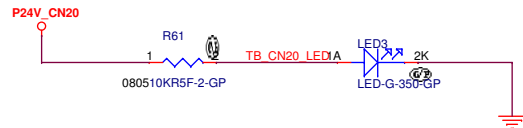
Power Output (CN18.CN19)



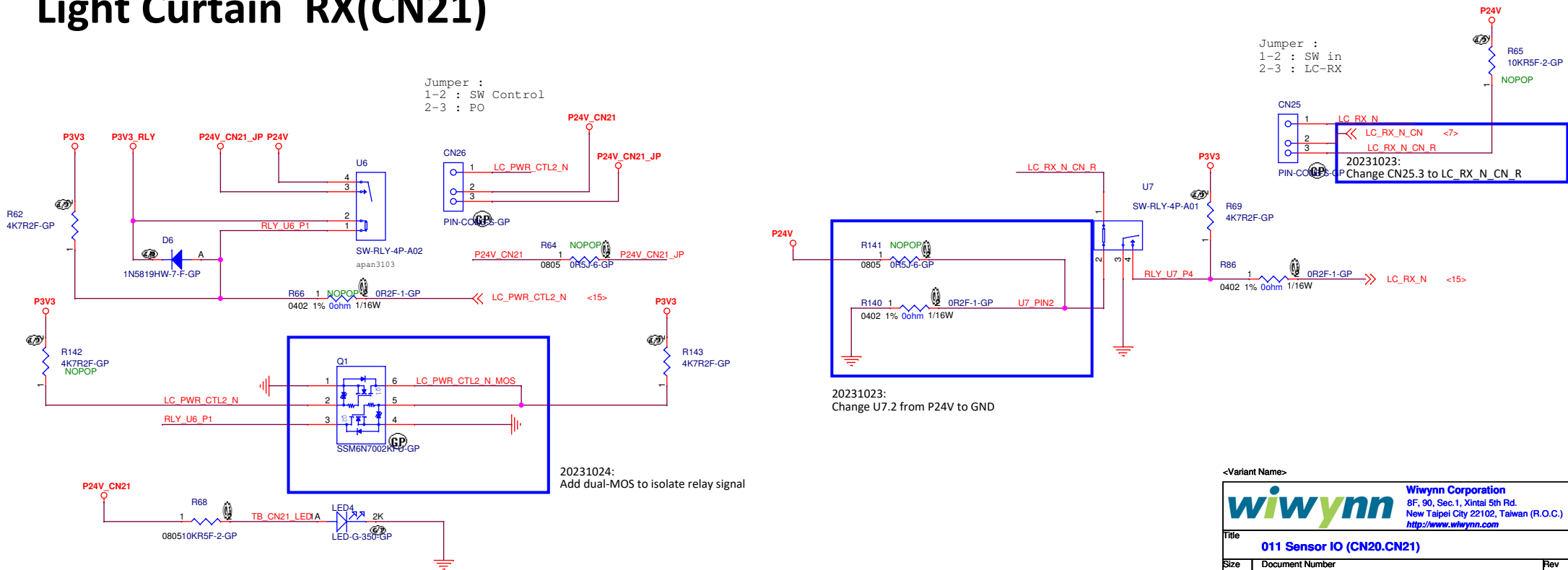
Light Curtain TX(CN20)



LED for validation



Light Curtain RX(CN21)



<Variant Name>

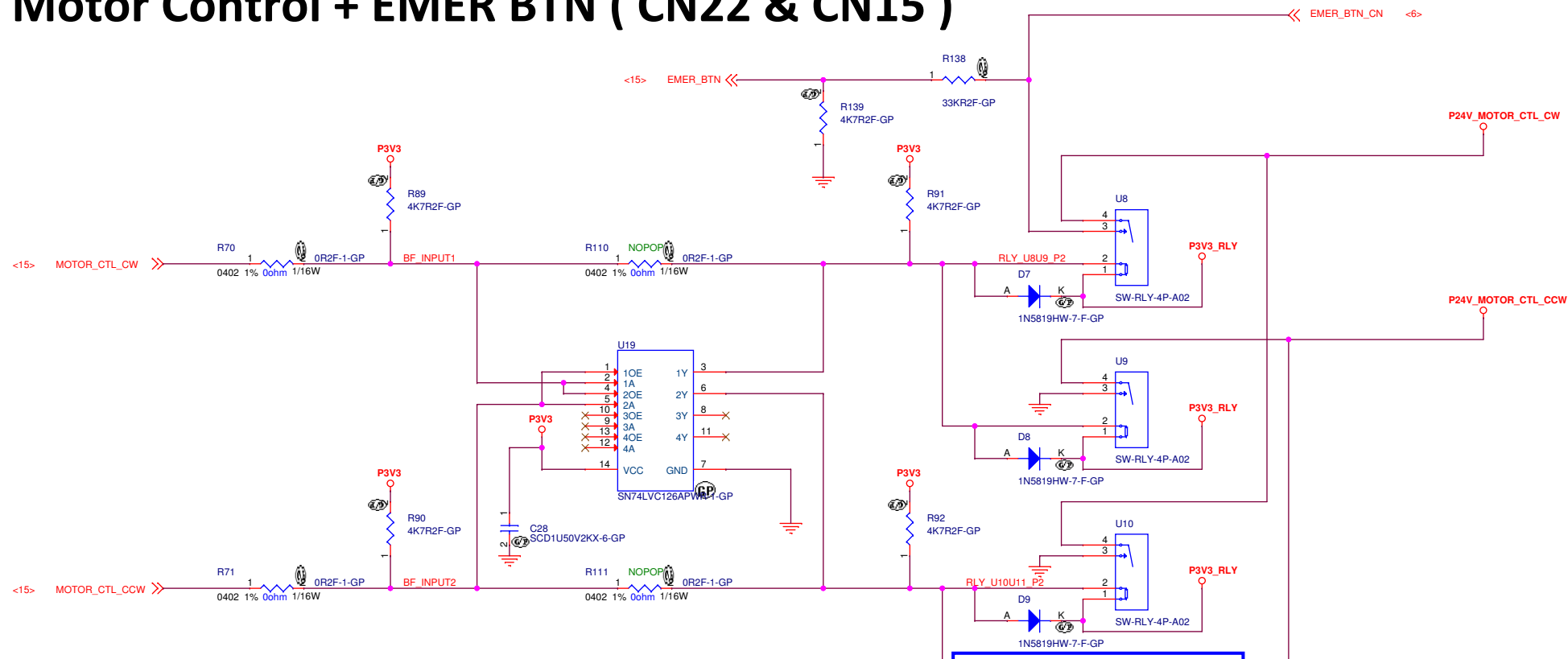
wiwynn Wiwynn Corporation
8F, 90, Sec.1, Xintai 5th Rd.
New Taipei City 22102, Taiwan (R.O.C.)
<http://www.wiwynn.com>

Title **011 Sensor IO (CN20,CN21)**

Size A3 Document Number **Sealion** Rev SA

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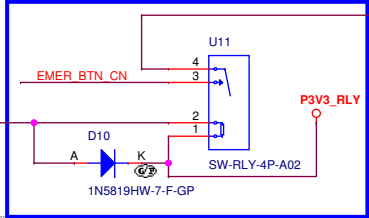
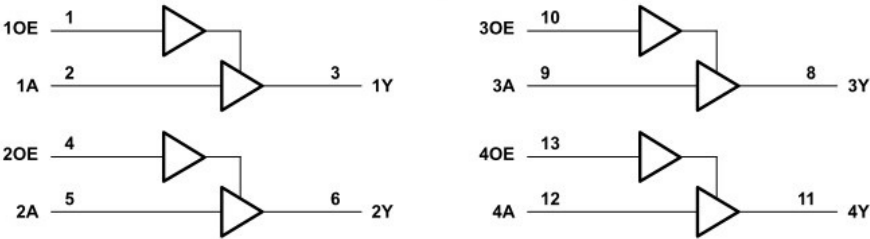
Door Motor Control + EMER BTN (CN22 & CN15)



FUNCTION TABLE
(EACH BUFFER)

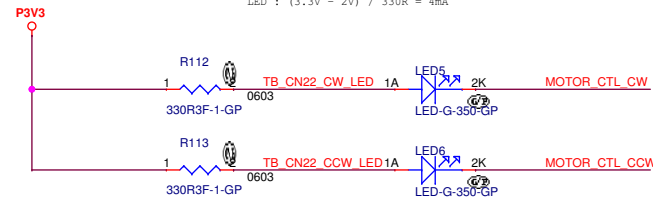
INPUTS		OUTPUT
OE	A	Y
H	H	H
H	L	L
L	X	Z

LOGIC DIAGRAM (POSITIVE LOGIC)

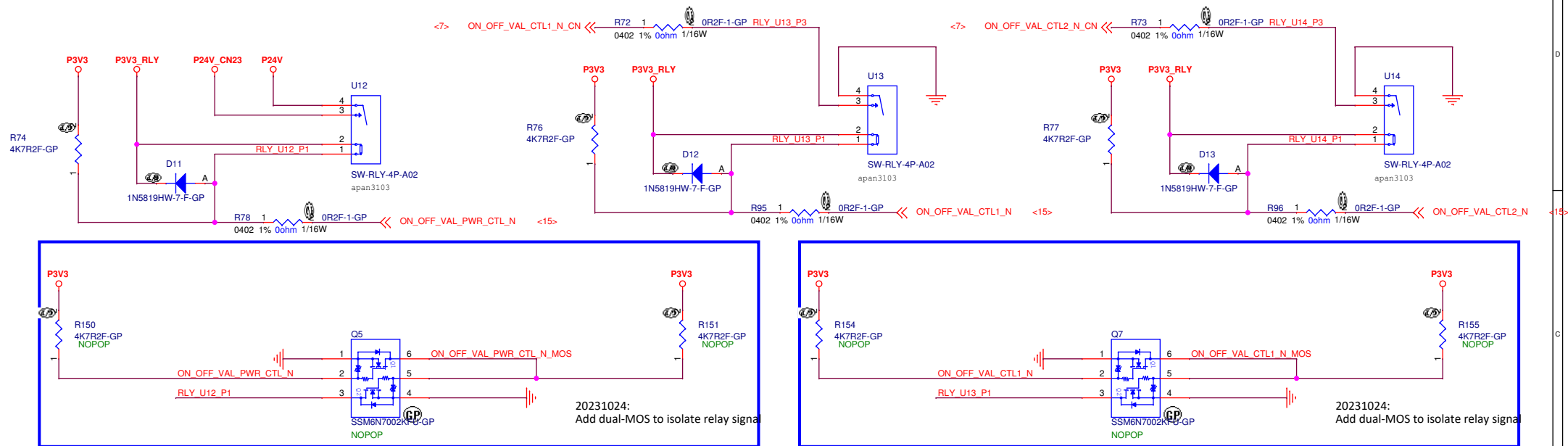


20231023:
Change U11.3 from P24V to EMER_BTN_CN

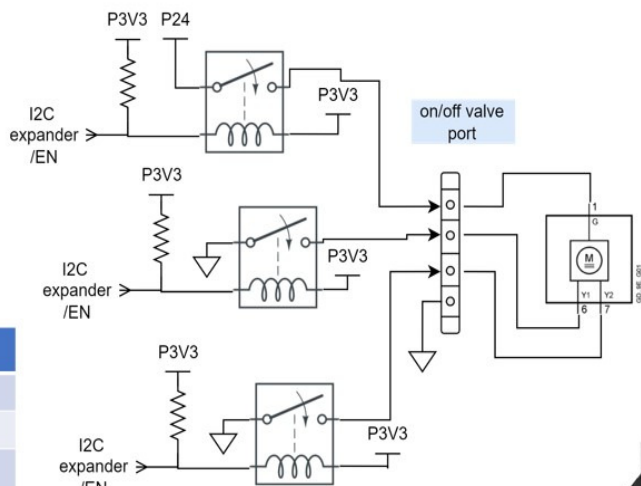
LED for Test



On-Off valve (CN23)



On-off valve



	P1	P2	P3
Y1向	0	0	1
Y2向	0	1	0
不轉	1	x	x
禁止	0	0	0

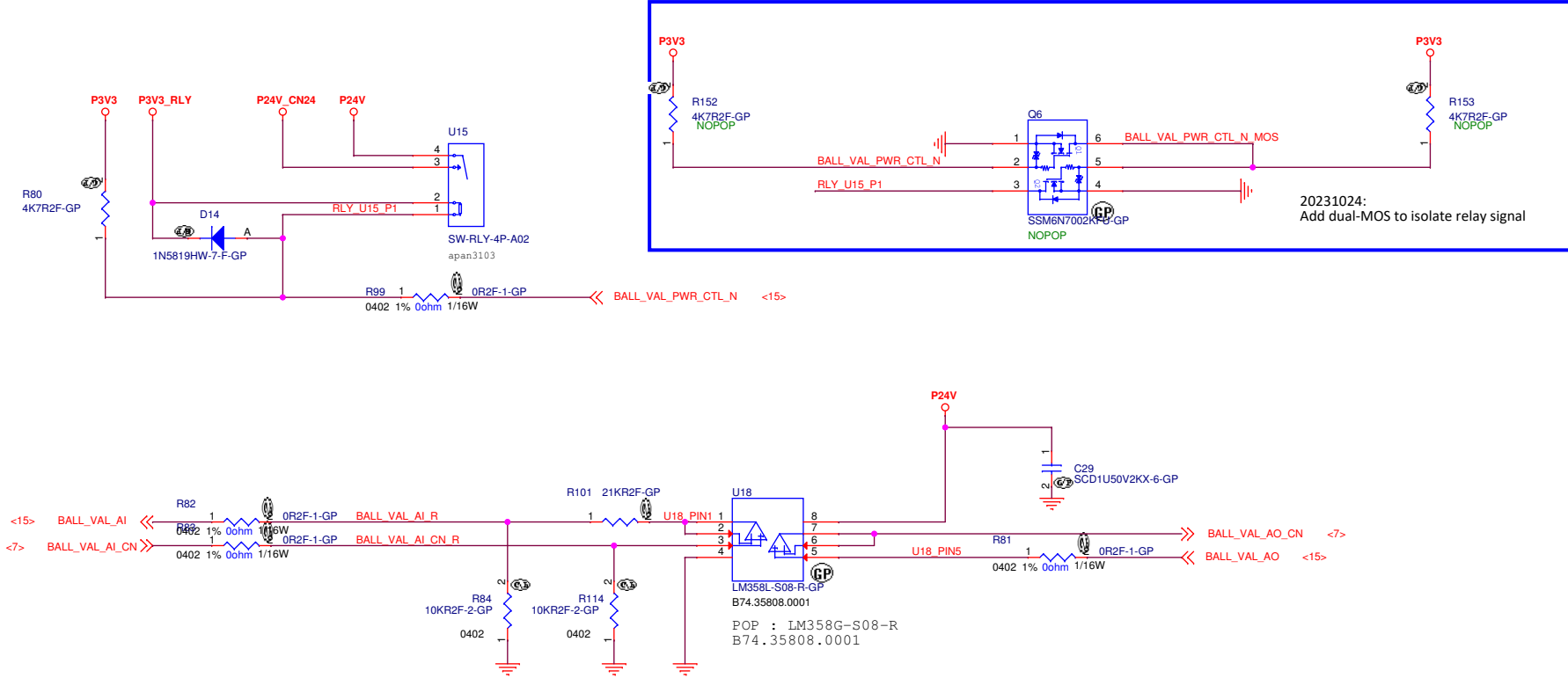


<Variant Name>

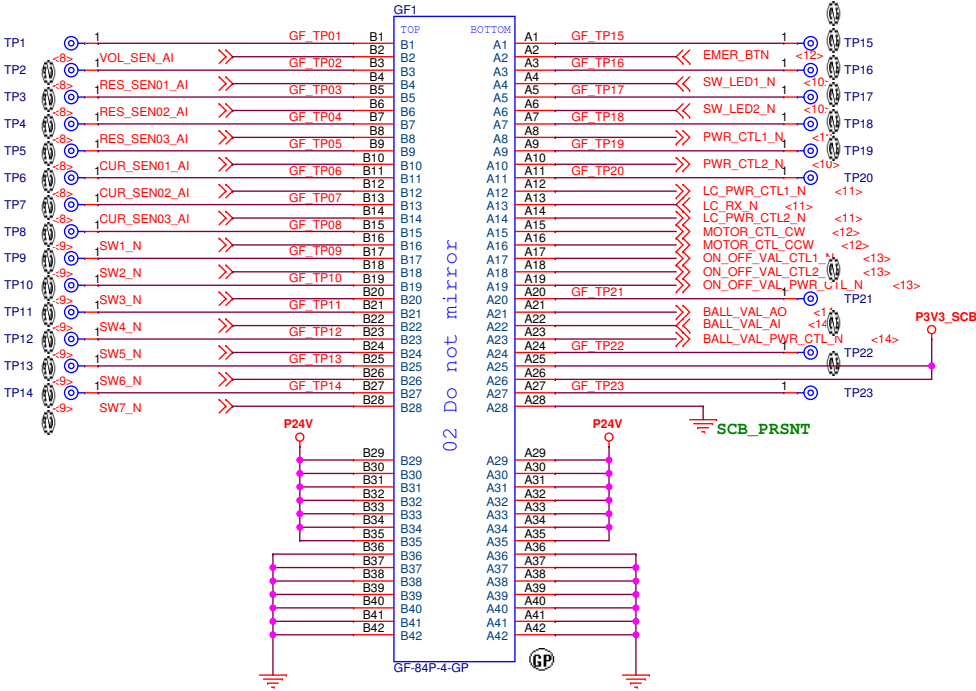
wiwynn Corporation
 8F, 90, Sec.1, Xintai 5th Rd.
 New Taipei City 22102, Taiwan (R.O.C.)
<http://www.wiwynn.com>

Title	013 Sensor IO (CN23)		
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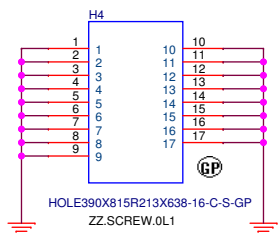
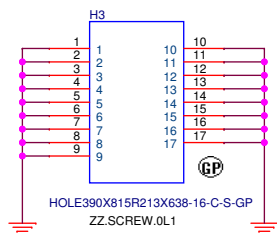
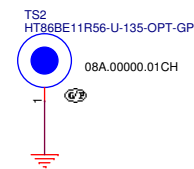
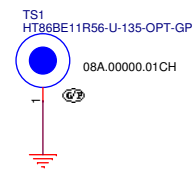
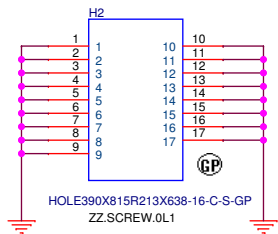
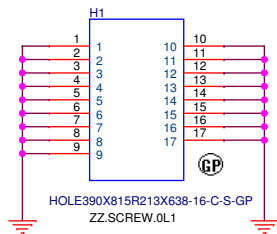
Rotary actuator for ball valves (CN24) only for NV2



GOLDEN FINGER

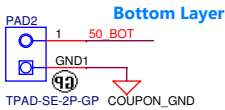
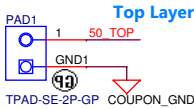


ME COMPONENT



IMPEDANCE COUPON

Single Ended : 50 ohm



CHANGE LIST

Version	Date	Item	Page	DESCRIPTION	BOM change	Layout change	Reason
SA	2023 06/07	1	8	Add 2*op amp for AI-V and AI-I sensor	✓	✓	To simplify schematic
SA	2023 06/12	2	8	R109, R127, R137 nonpop	✓		
SA	2023 06/12	3	11	Chagne U7_pin2 to 24V		✓	
SA	2023 06/12	4	12	Add emergency button function to turn off motor power directly	✓	✓	
SB	2023 10/23	5	12	U11.3 connected to EMER_BTN_CN		✓	For emergency button issue
SB	2023 10/23	6	8	Change U16 and U17 power from P24V to P3V3		✓	For voltage limit issue
SB	2023 10/23	7	11	Change U7.2 from P24V to GND		✓	For light curtain function
SB	2023 10/23	8	8	NOPOP R123	✓		Rsaved
SB	2023 10/23	9	11	Change CN25.3 to LC_RX_N_CN_R		✓	Light curtain function
SB	2023 10/24	10	11	Add dual-MOS to isolate relay signal		✓	Light curtain bypass function